

Trading and Exchanges — Detailed Summary (Larry Harris)

Source: Larry Harris, *Trading and Exchanges: Market Microstructure for Practitioners* (Oxford University Press, 2003) — read from the full text.

Overview & thesis

Harris's *Trading and Exchanges* is the canonical practitioner reference on **market microstructure** — "the branch of financial economics that investigates trading and the organization of markets" (Ch. 1, Introduction). It is deliberately *not* a book about how to value instruments: "This book is not about the securities and contracts that people trade... Books about investments and corporate finance examine these questions" (Ch. 1). Instead it explains the *business of trading* — who trades, why, how orders become executions, how prices come to reflect value, and what "liquidity" actually is.

The whole book is organized around explaining the origins of five characteristics of **market quality**: "Liquidity... Transaction costs... Informative prices... Volatility... Trading profits" (Ch. 1.2, Objectives). The secondary objective is to show "how market structure—trading rules and information systems—affect each of these five market characteristics."

The animating thesis is that **trading is a zero-sum game against the market average**: "Trading is a zero-sum game in which some traders win and others lose. Traders who do not expect to win should refrain from trading" (Ch. 1). Because "the combined gains and losses of buyers and sellers always sum to zero" (Ch. 8), "to trade profitably, traders must trade with people who will lose." The book's single most emphasized lesson, flagged as "the most important lesson you may learn from this book," is bleak for the amateur: "Uninformed traders lose simply because they trade. If you are an uninformed trader and do not want to lose, you should minimize your trading" (Ch. 14, Bid/Ask Spreads).

Two recurring engines drive everything: **information asymmetry** ("Well-informed traders profit at the expense of less informed traders," Ch. 1.5) and the **order-flow externality** ("Traders who offer to trade give other traders valuable options to trade for which they are not compensated... The order flow externality attracts and binds traders to markets," Ch. 1.5). The book is descriptive and institutional rather than mathematical — "I fully explain all essential concepts in the main text" (Ch. 1.1) — which is exactly why its vocabulary maps cleanly onto a screener.

How the book is organized

Harris tells you the plan explicitly: "The book is organized into seven parts" (Ch. 1.6). The read followed that arc:

- **Part I — The Structure of Trading** (Ch. 3–7): the trading industry (buy side/sell side), orders and their properties (Ch. 4), market structures (Ch. 5), order-driven markets and precedence rules (Ch. 6), brokers (Ch. 7).

- **Part II — The Benefits of Trade** (Ch. 8–9): *why* people trade and how markets benefit the economy.
- **Part III — Speculators** (Ch. 10–12): "the three main types of profit-motivated speculators" — informed traders, order anticipators, bluffers.
- **Part IV — Liquidity Suppliers** (Ch. 13–18): dealers, bid/ask spreads, block traders, value traders, arbitrageurs, and public order-submission strategies.
- **Part V — Origins of Liquidity and Volatility** (Ch. 19–20): the two synthesizing chapters.
- **Part VI — Evaluation and Prediction** (Ch. 21–22): transaction-cost measurement and performance evaluation.
- **Part VII — Market Structure** (Ch. 23–29): market-maker/specialist systems, competition among markets, bubbles and crashes, and insider trading.

Core concepts (deep)

Trader taxonomy (Ch. 8, Why People Trade). Harris uses two orthogonal cuts. By *motive*: "Our stylized traders are profit-motivated traders, utilitarian traders, or futile traders." Profit-motivated traders "trade only because they rationally expect to profit" (speculators and dealers). *Utilitarian traders* "trade because they expect to obtain some benefit from trading besides trading profits" — "investors, borrowers, asset exchangers, hedgers, and gamblers" (plus fledglings learning to trade, cross-subsidizers, and tax-avoiders). *Futile traders* "believe that they are profit-motivated traders... although they expect to trade profitably, their expectations are not rational." By *information*: "Traders are either informed traders or uninformed traders. Informed traders can form reliable opinions about whether instruments are fundamentally undervalued or overvalued." The buy side "buy[s] exchange services... Liquidity is the most important of these services"; the sell side (dealers, brokers, broker-dealers) sells them (Ch. 3).

Orders and liquidity (Ch. 4, Orders and Order Properties). "Orders are trade instructions... Your order submission strategy is the most important determinant of your success as a trader." The central duality: "An order offers liquidity—or equivalently supplies liquidity—if it gives other traders an opportunity to trade," while "traders who want to trade quickly demand liquidity" by taking standing offers. The "difference between the best ask and the best bid is the bid/ask spread... the inside spread," known also as "the touch" (UK) or "the vigorish" (betting).

Market structures (Ch. 5–6). Trading happens in **continuous** or **call** sessions, through three execution mechanisms: "In quote-driven systems, dealers arrange most trades... In order-driven systems, all trades are arranged by using order precedence rules to match buyers to sellers and trade pricing rules to determine the prices... In brokered trading systems, brokers arrange trades by helping buyers and sellers find each other" (Ch. 5). Order-driven markets run on "order precedence rules... and trade pricing rules" (Ch. 6). The primary rule is always **price priority**; the secondary rule is **time precedence** (futures) or **public-order precedence then time** (US stocks). The **minimum price increment (tick)** matters enormously: too small and it "weaken[s] the time precedence rule"; too large and "traders are reluctant to improve prices."

Dealers and inventory (Ch. 13, Dealers). "Dealers are merchants who make money by buying low and selling high." What they sell is immediacy: "The liquidity service they sell—immediacy—is valuable to

impatient traders." Crucially, "dealers are passive traders" who "trade when other traders want to trade," so they must "manage their inventories" and "be careful that they do not trade with better-informed traders." The **realized spread** is usually smaller than the quoted spread: "The realized spread is the difference between the prices at which dealers actually buy and sell," and in Harris's worked example a dealer quoting a 0.3 spread earns a **negative** realized spread of -0.1 after adverse news forces her quotes down.

Adverse selection and the spread (Ch. 14, Bid/Ask Spreads). "The bid/ask spread is the price impatient traders pay for immediacy." Harris decomposes it into **exactly two** components (a frequent point of confusion elsewhere): (1) the **transaction-cost (transitory) component**, which "compensates dealers for their normal costs of doing business" and "also funds any monopoly profits... and any risk premium that dealers may require for bearing inventory risk" — this component causes **bid/ask bounce**; and (2) the **adverse-selection component**, "that part of the bid/ask spread that compensates dealers for the losses that they suffer when trading with well-informed traders." Inventory risk lives *inside the first component*, not as a separate third piece. Under free entry, "dealers just earn normal profits" (economic profit $\rightarrow$ zero).

Order anticipators and bluffers (Ch. 11–12). Not all speculators make prices informative. **Order anticipators** (front runners, sentiment-oriented technical traders, squeezers) "are parasitic traders. They profit only when they can prey on other traders. They do not make prices more informative, and they do not make markets more liquid" (Ch. 11). **Bluffers** "try to fool other traders into trading unwisely" via rumormongering or price manipulation ("paint the tape") (Ch. 12).

Block traders (Ch. 15). "Block trades result from orders that are too large to fill easily using standard trading procedures." They migrate to the "upstairs block market," where block initiators meet block liquidity suppliers; the initiator's chief risk is exposing size to "front runners and quote matchers."

Value traders and the winner's curse (Ch. 16). "Value traders... are the ultimate suppliers of market liquidity. They trade when no one else will." But "value trading is profitable only when price differs from fundamental value" — i.e., "only when value traders trade in response to demands for liquidity made by uninformed traders." **Market resiliency** is their gift: "Value traders make markets resilient by standing ready to trade when prices [move] away from fundamental values" (Ch. 16.1.3). Their two risks are "adverse selection and the winner's curse" — the latter being the regret of having "misestimated instrument values."

Arbitrage (Ch. 17). "Arbitrageurs are speculators who trade on information about relative values," enforcing "the law of one price." Besides being informed traders they "supply liquidity, move liquidity, and produce financial products."

Liquidity as bilateral search (Ch. 19). "Liquidity is the ability to trade large size quickly at low cost when you want to trade. It is the most important characteristic of well-functioning markets." Harris warns "liquidity means different things to different people" precisely because it is multi-dimensional, and it is "the successful outcome of a bilateral search in which buyers look for sellers and sellers look for buyers." Exchanges exist to cut that search cost: "Exchanges Are Search Engines."

Volatility (Ch. 20). "Volatility is the tendency for prices to unexpectedly change," split into "fundamental volatility... due to unanticipated changes in instrument values" and "transitory volatility... due to trading activity by uninformed traders." The tell: fundamental volatility shows up as *unpredictable* price changes — "an unpredictable price process is called a random walk" — while "transitory volatility is closely related to the transaction costs that uninformed traders bear."

Market efficiency (Ch. 10). "A price is informative when it is near its corresponding fundamental value." "Changes in fundamental values are completely unpredictable," so efficient-market price changes approximate a random walk. Fischer Black's cited yardstick: prices are "informative if they are between one-half and twice their fundamental values." Markets out-forecast individuals because "markets aggregate data from many sources."

Key frameworks, metrics & formulas

- **The 2×2 trader map:** {profit-motivated | utilitarian | futile} × {informed | uninformed} (Ch. 8). This is the lens for reading *who is on the other side of your trade*.
- **Three execution systems:** quote-driven (dealers) · order-driven (precedence rules) · brokered (search) — most real venues are hybrids (Ch. 5).
- **Order-precedence stack:** price priority (primary) → time precedence / public-order precedence (secondary), governed by tick size (Ch. 6).
- **Two spread components:** total spread = transaction-cost/transitory component (normal costs + monopoly profit + inventory-risk premium; drives bid/ask bounce) + adverse-selection component (Ch. 14). "To actually estimate the two spread components, analysts must use econometric methods."
- **Quoted vs. realized (effective) spread:** realized = prices dealers *actually* buy and sell at ≤ quoted, because of price improvement and quote adjustment (Ch. 13).
- **Four dimensions of liquidity (Ch. 19):** *immediacy* (speed — "one of several dimensions of liquidity," Ch. 4/19), *width* (the bid/ask spread — the price of immediacy, Ch. 14), *depth* (size available at/near the quote), and *resiliency* (speed of price recovery after uninformed order imbalance, Ch. 16.1.3).
- **Three transaction-cost categories (Ch. 21):** *explicit* (commissions, fees, taxes — "costs a cost accountant would easily identify") · *implicit* (spread paid + price impact) · *missed-trade opportunity* costs (orders that don't fill).
- **Price-benchmark cost estimator (Ch. 21.2), the book's core formula:** Estimated Cost = $\text{TradeSize} \times \text{TradeSign} \times (\text{TradePrice} - \text{BenchmarkPrice})$ where TradeSign = +1 for a buy, −1 for a sale. Corollary: "Transaction cost to one side is trading profit to the other side" — estimated costs across both sides sum to zero. Harris distinguishes these specified-benchmark methods (best for specific trades) from *econometric* methods (best for whole-market averages).
- **Two volatility components (Ch. 20):** fundamental (random-walk, unforecastable) vs. transitory (mean-reverting, tied to uninformed trading and transaction costs).
- **The blunt performance prior (Part VI):** "Few profit-motivated traders consistently outperform the market. Most active traders lose because they trade too much and because they pay too much to trade."

Chapter / thematic walkthrough

- **Ch. 1–2 (Introduction, Trading Stories):** frames the five market-quality objectives, the zero-sum game, and "trading is a search problem"; Ch. 2 grounds it in narrated retail, block, futures, and FX trades.

- **Ch. 3–7 (Structure of Trading):** industry map (buy/sell side; dealers vs. brokers vs. broker-dealers), the offer/take-liquidity duality of orders, continuous vs. call sessions and the three execution systems, precedence rules and tick size, and the broker principal-agent problem.
- **Ch. 8–9 (Why People Trade):** the trader taxonomy; the insistence that you must know your *own* reason for trading, and that volume interpretation depends on knowing *why* others trade.
- **Ch. 10 (Informed Traders and Market Efficiency):** fundamental vs. market value, noise, random walks, information aggregation — the theory of *how prices become informative*.
- **Ch. 11–12 (Order Anticipators, Bluffers):** the parasitic and manipulative speculators — front-running, squeezes, rumormongering, painting the tape.
- **Ch. 13–14 (Dealers, Bid/Ask Spreads):** how dealers quote, manage inventory, and set spreads; the two-component spread; the "minimize your trading" lesson.
- **Ch. 15–17 (Block Traders, Value Traders, Arbitrage):** where large size finds liquidity; value traders as the deepest liquidity supply and source of resiliency; arbitrage as connective tissue across markets. Ch. 18 adds when public limit orders displace dealers.
- **Ch. 19–20 (Liquidity, Volatility):** the synthesizing chapters — liquidity as bilateral search with multiple dimensions; volatility split into fundamental vs. transitory.
- **Ch. 21–22 (Transaction Cost & Performance):** the cost taxonomy and price-benchmark estimator; why implementation cost usually dominates selection skill.
- **Ch. 23–29 (Market Structure):** specialist/market-maker systems, competition within/among markets, bubbles/crashes/circuit breakers, and insider trading (Ch. 29 — "material, nonpublic information," the SEC definition, the Texas Gulf Sulfur case).

How it maps to the Trader Screener

1. **§2c Average daily volume (ADV)** and **§2c Dollar volume (ADV × price)** are the honest free stand-in for the missing quote feed. Ch. 21 says implicit costs (spread + price impact) blow up with size and in illiquid names — "the price impacts of implementing them in large size may cause them to lose on net." Dollar volume is the [computable] "can I get size in and out?" filter that proxies Harris's *depth* dimension (Ch. 19) without a book.
2. **§2c Bid/ask spread** is Harris's *width* dimension — literally "the price impatient traders pay for immediacy" (Ch. 14). It is correctly marked [paid] (needs a quote feed); documenting it as out-of-v1 tells the build exactly which liquidity dimension it is *not* measuring, with the Ch. 21 price-benchmark formula as the reference for any later effective-cost work.
3. **§2c Market depth / book** is Harris's *depth* dimension and, by extension, the only window onto *resiliency* (Ch. 16.1.3, 19). Both are [paid] real-time-only; resiliency in particular cannot be measured from nightly OHLCV at all — a limit worth stating explicitly.
4. **§2e Congress trades**, **§2e Insider trades (EDGAR)**, **§2e Committee-conflict cross-tab**, and **§2e News sentiment** are the **adverse-selection component made observable** (Ch. 10, 14, 29). Harris's informed-trader theory is the moat's theoretical basis: these [free] free signals flag the very order flow that widens spreads because "uninformed traders lose to well-informed traders through the intermediation of

dealers" (Ch. 1). This deserves its own scored view, not just raw event flags.

5. **§2b Realized (historical) volatility** and **§2f VIX** operationalize Ch. 20's fundamental-vs-transitory split — regime context for *when* liquidity thins and transaction costs spike. **§2b Beta (vs SPY)** and **§2b ATR / ATR%** are the practical realized-vol surfaces the book's volatility framework justifies.

6. **§2a Return 1/3/6/12-month** and **§2a Relative strength rank (vs universe)** are price-discovery signals (Ch. 10): informed buying/selling drags price toward fundamental value, which shows up as momentum. But per Ch. 14/21 they must be **liquidity-aware** — a momentum rank is only tradeable if §2c dollar volume clears a floor, else price impact eats the edge.

7. **§2g Suggested position size (ATR-based)** and **§2g Stop distance (ATR multiple)** are the discipline Ch. 21 demands: since "most active traders lose because they trade too much and... pay too much to trade," ATR-scaled sizing caps price impact per name. **§2a VWAP** ([paid] if real-time) is precisely the kind of benchmark price the Ch. 21 estimator compares fills against.

8. **§2d Composite score (model.json)** and **§2d Upside % (DCF/comps)** are the screener's *value-trader* engine (Ch. 16): flag where price diverges from estimated fundamental value. The §2e smart-money signals then act as the *timing/confirmation* overlay — value trading "is profitable only when value traders trade in response to demands for liquidity made by uninformed traders."

Actionable takeaways

- Ship **§2c Dollar volume (ADV × price)** as a P1 default column *and* a hard liquidity filter — the highest-value [computable] metric and the honest substitute for the absent quote feed (Ch. 19, 21).
- Give the **§2e smart-money block its own dedicated view/filter**: Harris's informed-trader theory (Ch. 10, 14) makes it the screener's most defensible edge — an observable proxy for the adverse-selection flow professionals fear.
- Label [paid] **§2c Bid/ask spread** and **§2c Market depth / book** "not in v1," citing the Ch. 21 price-benchmark method as the future reference, so the liquidity-dimension gap is explicit rather than silent.
- Treat momentum (**§2a**) and value (**§2d**) as *liquidity-gated*: rank on them, but never surface a name that fails the §2c dollar-volume floor — the Ch. 14/21 price-impact lesson.
- Use **§2b realized vol / §2f VIX** as a regime switch (Ch. 20), not a stock picker: they tell you when transitory volatility and transaction costs are elevated market-wide.
- Correct the record in any doc: Harris's spread has **two** components (transaction-cost + adverse-selection), with inventory risk *inside* the first — not a three-way split.
- For a swing/position horizon, the free nightly [free]+[computable] pipeline covers the vast majority of what the book says matters; only day trading forces the [paid] intraday feeds (spread, depth, VWAP).

Caveats, limits & what to distrust

- **Vintage.** The text is the 2002/2003 edition; Harris himself warns "some specific descriptive information in this book will undoubtedly be dated... The economic principles governing markets... however, will remain the same" (Ch. 1.8). Treat the *institutions* (decimalization debates, specialist floors, ECNs) as historical; trust the *economics*.
- **Descriptive, not predictive-quantitative.** The book gives frameworks and one core cost formula, not a fitted alpha model. It will not tell you a dollar-volume floor or an optimal momentum lookback — those are empirical choices it only frames.
- **Spread decomposition is an estimation problem.** "To actually estimate the two spread components, analysts must use econometric methods" (Ch. 14). Do not expect to back out adverse-selection cost from a screener; treat §2e as a *flag*, not a calibrated cost.
- **The screener has no quote/trade tape**, so three of the four liquidity dimensions (width/spread, depth, resiliency) are unmeasurable in v1. Dollar volume is a proxy for *depth only* — it says nothing about spread or resiliency.
- **Zero-sum framing is against the market average**, not absolute (Ch. 8). It is a discipline device ("minimize your trading"), not a claim that buy-and-hold investors lose money.
- **Survivorship in the smart-money moat:** Congress/insider signals are noisy and lagged by filing rules; Harris's own insider-trading chapter (Ch. 29) stresses how hard even regulators find it to identify informed trading cleanly.

Open questions

- Would a daily-data **effective-cost estimate** (a crude Ch. 21 price-benchmark proxy) add signal over raw §2c dollar volume, or not enough to justify a [computable] column?
- What **dollar-volume floor** cleanly separates "tradeable" from "cost-prohibitive" for the intended position size?
- Should **§2e adverse-selection intensity** become its own scored dimension (weighting congress + insider + committee-conflict + sentiment) rather than raw event flags?
- How should the screener **combine value (§2d) and momentum (§2a)** given Harris's point that value trades pay off only when uninformed flow has pushed price away from value — is there a §2e-based trigger that times the entry?
- Is there a lightweight **regime gate** (§2f VIX / rates) that should suppress or resize signals when transitory volatility is high (Ch. 20)?

Sources

- **Primary:** Larry Harris, *Trading and Exchanges: Market Microstructure for Practitioners* (Oxford University Press, 2003) — **summarized from the full text.**
- **Chapters referenced:** Ch. 1 (Introduction / objectives, key themes, zero-sum game) · Ch. 3 (The Trading Industry / buy side · sell side) · Ch. 4 (Orders and Order Properties) · Ch. 5 (Market Structures) ·

Ch. 6 (Order-driven Markets / precedence rules, tick size) · Ch. 7 (Brokers) · Ch. 8 (Why People Trade / trader taxonomy) · Ch. 10 (Informed Traders and Market Efficiency) · Ch. 11 (Order Anticipators) · Ch. 12 (Bluffers and Market Manipulation) · Ch. 13 (Dealers / realized spread) · Ch. 14 (Bid/Ask Spreads / two components) · Ch. 15 (Block Traders) · Ch. 16 (Value Traders / winner's curse, resiliency §16.1.3) · Ch. 17 (Arbitrage) · Ch. 19 (Liquidity / bilateral search, dimensions) · Ch. 20 (Volatility / fundamental vs. transitory) · Ch. 21 (Liquidity and Transaction Cost Measurement / price-benchmark formula) · Ch. 22 (Performance Evaluation) · Ch. 29 (Insider Trading).